Magenta gap: a geometric reachability model of nonspectral hue and color gamut boundaries

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Abstract

Monochromatic stimuli are ordered by wavelength, whereas hue includes a nonspectral purple closure. We model this closure by inserting a purple interval into a compactified spectral arc, then incorporate the resulting hue circle into a color body with an achromatic quotient. For a prescribed boundary and an explicitly calibrated indefinite metric, nullity bounds the angular variation of the saturation ceiling; a diagonal construction requires strict inequality. The symmetric body also has a standard two-level effect representation, without identifying effect magnitude with opponent contrast. A separate bounded-reflectance signal model supplies a certified boundary point with a unique tangent plane containing the achromatic direction. At that point no Lorentzian metric making this direction timelike can make the boundary null, even with cross-terms. The results separate conditional geometric compatibility from the encoding, calibration and controls required for a perceptual realization.

1 Introduction

Color perception provides a concrete setting for asking how physical conditions give rise to an organized field of appearance. A theory of color must relate spectra, color matches, hue relations, discrimination, and changes with viewing conditions. We call the task of specifying those relations the *transformation problem* for color appearance. Its object is the organization of color experience and its dependence on physical and physiological conditions, rather than an isolated experience called "magenta." The problem can be stated before a particular geometric or neural account is complete.

Magenta makes one distinction especially clear. Wavelength orders monochromatic stimuli along a line, whereas hue includes a route from violet to red through the nonspectral purples. Magenta has no single-wavelength correlate, although physically ordinary mixtures induce it. The full physical stimulus domain already contains those mixtures: the wavelength interval is only a restricted locus within it. There is consequently no interval-to-circle obstruction to a physical account of color. The "magenta gap" names the absence of a monochromatic representative and motivates a response-side description of hue closure.

Standard trichromatic colorimetry already accounts for the closure of the realizable chromaticity gamut through nonspectral mixtures. Positive mixtures give normalized cone responses in the convex hull of normalized spectral responses; the familiar chromaticity boundary includes the purple

line [1]. This supplies a physical mixture geometry, but does not by itself identify that boundary with a calibrated circle of perceived hues. Which directions correspond to which hues, how they are spaced, and how saturation and discrimination are measured require an appearance model. The distinction concerns what a given model specifies, not a claim that color science lacks accounts of these features.

Geometrical approaches to color have a substantial history, including Schrödinger's work on color metrics [2]. Color-matching structure is formalized by Grassmann representations [3]; differential-geometric models address additional questions about color-space organization and discrimination [4]. Appearance also depends on adaptation, surround, and the viewing regime [5, 6]. The explanatory task therefore includes the observer and context, rather than treating the focal spectrum alone as a complete description. Throughout, we distinguish the stimulus class and its signal image, a proposed organization of appearance, and the subset of that appearance space reached under specified conditions. The maps relating these objects carry the empirical content.

The relation between color science and color ontology has long been contested. Hardin places the organization of color experience in dialogue with psychophysics and visual physiology [7]; Thompson develops an ecological account in which colors depend jointly on perceiver and environment [8]; Byrne and Hilbert defend colors as physical properties, specifically types of reflectance [9]. The geometric questions pursued here can be separated from those competing ontologies: what structure represents the reported organization of appearances, and what maps connect it to stimuli, observer states, and viewing conditions?

Two-level effect models already connect color solids with positive cones and Lorentzian algebra [10, 11]. We use their effect-body geometry as an algebraic comparison, not as a new derivation of that geometry or of a physiological response map. Separately, the classical object-color solid concerns bounded reflectances under a fixed illuminant [12, 13]. These two constructions describe different objects until an encoding relates them.

This paper develops a geometric model in three stages. First, the ordered spectral arc is completed by a purple interval to form a resolved hue circle. Second, hue is collapsed at zero chromatic radius, giving a three-dimensional color body with an achromatic axis. Third, an indefinite metric is introduced as a *reachability gauge*: a device for representing a prescribed boundary of ordinary optical access. The metric is not a perceptual-distance function, and its signature does not follow from the topology. It is a further modeling assumption whose mathematical consequences can be stated separately from its physiological interpretation.

The principal result concerns a positive hue-dependent ceiling $\alpha(\theta)$ and the boundary

$$R(\theta, z) = \alpha(\theta)(T - |z|). \tag{1}$$

If each open sheet is null for the calibrated metric specified below, with positive-definite spatial block, then

$$\left| \frac{\mathrm{d} \log \alpha}{\mathrm{d}\theta} \right| \leq \beta(\theta, z), \tag{2}$$

where β is the boundary angular metric scale divided by its chromatic radius. The inequality is covariant under reparameterization of hue. A finite positive radial coefficient in the diagonal construction requires strict inequality. These statements constrain the compatibility of a boundary and a chosen metric; they do not independently establish either as a description of human vision.

The distinction is consequential. The bicone shape and its calibration are inputs, not predictions derived from cone physiology. Its equatorial ridge follows from the assumed symmetric shape. A related obstruction to a smooth null radial envelope holds under explicit metric and boundary hypotheses. Neither statement proves that optical response trajectories are confined by the metric.

We therefore separate the geometry, a numerical consistency calculation using cone fundamentals, and the additional control condition that a reachability interpretation would require.

Section 6 makes the limitation concrete. In a specified continuum object-color model, an exact-arithmetic calculation certifies a differentiable boundary point whose tangent plane contains the achromatic direction. A short Lorentzian argument rules out nullity there whenever that direction is timelike. This excludes one signal-boundary identification; it neither changes the conditional bicone theorem nor identifies a perceptual boundary with the signal solid.

2 Spectral stimuli and resolved hue

2.1 Stimulus space and nonspectrality

Let a denote an observer and fixed viewing regime, including the adaptation and surround conditions relevant to the response. Let $\mathcal{S}_a$ be an admissible set of finite positive spectral measures, and let

$$\Phi_a : \mathcal{S}_a \longrightarrow M_a \tag{3}$$

be its response map. An additive-light model may take $\mathcal{S}_a$ to be a cone; bounded reflectances under fixed illumination instead give a bounded stimulus class. Measures accommodate both continuous spectral power distributions and idealized monochromatic stimuli. In the additive model the latter have the form $t\delta_\lambda$, where $t > 0$ specifies intensity. After choosing an intensity normalization, the wavelength interval

$$I_a = (\lambda_-^a, \lambda_+^a) \tag{4}$$

parametrizes a one-dimensional stimulus locus. The endpoints are ideal limiting directions of the chosen model; finite numerical cutoffs are experimental or computational proxies.

The map Φ_a induces a metameric relation,

$$\mu \sim_a \nu \quad \Longleftrightarrow \quad \Phi_a(\mu) = \Phi_a(\nu). \tag{5}$$

A response class is spectral when it contains a monochromatic representative $t\delta_\lambda$ under the stated regime and permitted intensity range. A nonspectral class contains none. This is a condition on the preimage of the response map. It does not deny the physical causation of nonspectral responses, and it does not make the entire stimulus space one-dimensional. For a hue-only classification, the same definition is applied after the appropriate appearance coordinates have been fixed or projected out.

Human purple responses are paradigmatically induced by mixtures with short- and long-wavelength contributions. A mixture coordinate is consequently a response-side or colorimetric coordinate, not a wavelength between red and violet. The actual relation between mixture weights and hue depends on the observer and regime. The construction below records an ordered purple continuum without claiming that its parameter is perceptually uniform or directly proportional to radiant power.

Honeybee vision makes this observer dependence concrete. Its ultraviolet-, blue-, and green-sensitive receptors sample spectra differently from human cones. A nonspectral UV–green mixture sector is conventionally called "bee-purple" [14]; Chittka's color-hexagon construction plots mixtures of ultraviolet and green monochromatic lights alongside the spectral locus [15]. Verhoeven, Ren, and Lunau visualize floral patterns with UV, blue, and green photographs, assigning those channels to human-visible display colors [14]. These false colors represent differences in receptor stimulation, not a reproduction of a bee's subjective experience. The comparison supports an observer-relative account of stimulus encoding and nonspectral mixtures. It does not establish that bee and human

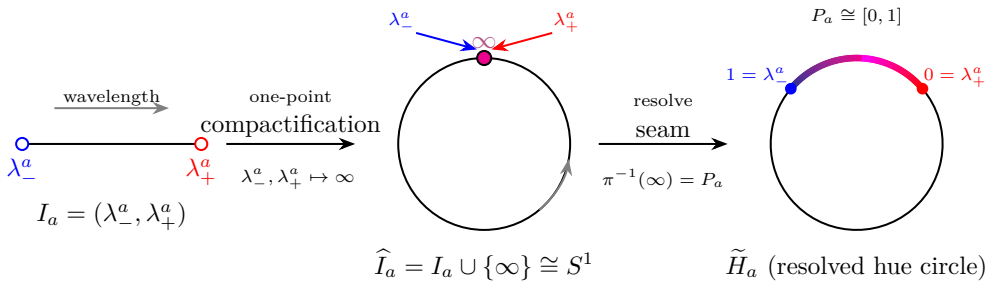


Figure 1: Compactification and resolution of the modeled hue circle. The physical interval, its abstract one-point compactification, and the resolved response circle are distinct objects. Inserting P_a represents the purple continuum without adding wavelengths. The closed interval includes the two attachment classes; its interior is the nonspectral arc. This is a structural model of hue organization, not a physiological derivation.

phenomenal hue spaces have the same topology. Applying the present circle construction to bees would require a specified response map and behavioral interpretation, rather than an inference from receptor count alone.

We idealize the spectral hue responses used in the construction as an ordered arc. This is an explicit modeling restriction: the full spectral-response curve and its hue projection need not be globally injective in every regime. The interval gluing describes the intended hue organization once that arc and its limiting classes have been identified. It is not a theorem that all visual systems must implement this particular construction.

2.2 Compactification and insertion of the purple interval

Since an open bounded interval is homeomorphic to $\mathbb{R}$, its one-point compactification is a circle,

$$H_a = \widehat{I}_a = I_a \cup \{\infty\} \cong S^1. \tag{6}$$

The added point is an abstract closure marker. It is neither a wavelength nor an additional stimulus. Equivalently, $\widehat{I}_a$ is obtained from the auxiliary closed interval $\overline{I}_a$ by identifying its endpoints. This identifies the ends only in the auxiliary topological model; it does not identify the physical spectral directions.

To represent distinguishable purples, introduce $P_a = [0,1]$ and glue its endpoints to those of $\overline{I}_a$:

$$\widetilde{H}_a = \overline{I}_a \cup_{\partial I_a} P_a \cong S^1, \qquad 0 \sim \lambda^a_+, \quad 1 \sim \lambda^a_-. \tag{7}$$

Thus $\tau = 0$ denotes the red-side limit and $\tau = 1$ the violet-side limit of the purple interval. The spectral arc runs in the opposite direction, closing the circle. A collapse map records this insertion:

$$\pi : \widetilde{H}_a \longrightarrow H_a, \qquad \pi^{-1}(\infty) = P_a. \tag{8}$$

On the spectral interior, π identifies the two copies of I_a. The endpoints of P_a are attachment classes; its interior represents the nonspectral transition. Figure 1 summarizes the construction.

The word "resolution" refers here to inserting an interval over the abstract seam. The compactification point is already a regular point of a circle; there is no mathematical singularity requiring a blow-up. The operation gives a way to retain the distinction between spectral and purple arcs within a common cyclic target space. Both arcs belong to the response model, but only the spectral arc is supplied with a wavelength parameter.

A smooth structure may be chosen on $\widetilde{H}_a$. Its existence does not imply that the raw wavelength coordinate and a chosen mixture coordinate have matching derivatives at the seams. Smooth charts require suitable local reparameterizations. The model therefore separates the topology of the glued circle from its angular calibration,

$$\theta : \widetilde{H}_a \longrightarrow \mathbb{R}/(2\pi\mathbb{Z}). \tag{9}$$

Changing this calibration transforms the metric components and derivatives used below. There is no preferred angular spacing determined by the gluing alone.

3 Quotient color body and metric model

3.1 The achromatic quotient

Let $r \in [0, r_{\max}]$ denote a chromatic radius and $z \in L_a$ a centered height coordinate, provisionally called luminosity. Its relation to opponent activation, luminance or brightness requires calibration. Neither coordinate is assumed at this stage to be perceptually uniform. Hue is undefined at $r = 0$, so the ambient color body is modeled by

$$M_a = \big([0, r_{\max}] \times \widetilde{H}_a \times L_a\big)/\sim, \qquad (0, h, z) \sim (0, h', z). \tag{10}$$

The achromatic axis is $A_a = \{r = 0\}$. Its points carry luminosity but no chromatic phase. On $M_a^* = M_a \setminus A_a$, the projection to $\widetilde{H}_a$ has local fiber coordinates (r, z). This bundle description applies only to the chromatic stratum. Treating the entire body as a product with the hue circle would artificially assign every neutral state a family of hue labels.

Figure 2 displays this quotient separately from the proposed ordinary preparation image. At every height, all zero-radius hue labels represent the same achromatic point; away from the axis, the resolved hue coordinate remains available.

Choosing $x = r\cos\theta$ and $y = r\sin\theta$ identifies the chromatic disk in ordinary Cartesian coordinates. A convenient baseline metric is

$$g_0 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 - \kappa^2\mathrm{d}z^2 = \mathrm{d}x^2 + \mathrm{d}y^2 - \kappa^2\mathrm{d}z^2, \qquad \kappa > 0. \tag{11}$$

The polar chart fails on the axis, but the Cartesian tensor is regular there. The period 2π is part of this coordinate convention. Under a nonlinear angular reparameterization, the component multiplying the new angular differential must transform; one cannot preserve the bare r^2 expression and simultaneously claim to have kept the same metric.

The gauge and a local discrimination metric are different objects. Choosing a regular Cartesian core for g_0 does not specify discrimination thresholds near gray, nor does it identify coordinate circumference with perceptual circumference. Indow's multidimensional-scaling studies of the Munsell solid reconstruct perceptual relations from judgments of color difference [16]; a chosen angular period alone does not establish that empirical geometry. Empirical anisotropy may enter a separately calibrated discrimination tensor. The boundary identification used in Sec. 5.1 is an additional assumption linking that tensor to the reachability construction.

3.2 Fixed-hue sections and the role of the signature

Each fixed-hue radial–luminosity section has the quadratic form

$$g_\theta = \mathrm{d}r^2 - \kappa^2\mathrm{d}z^2. \tag{12}$$

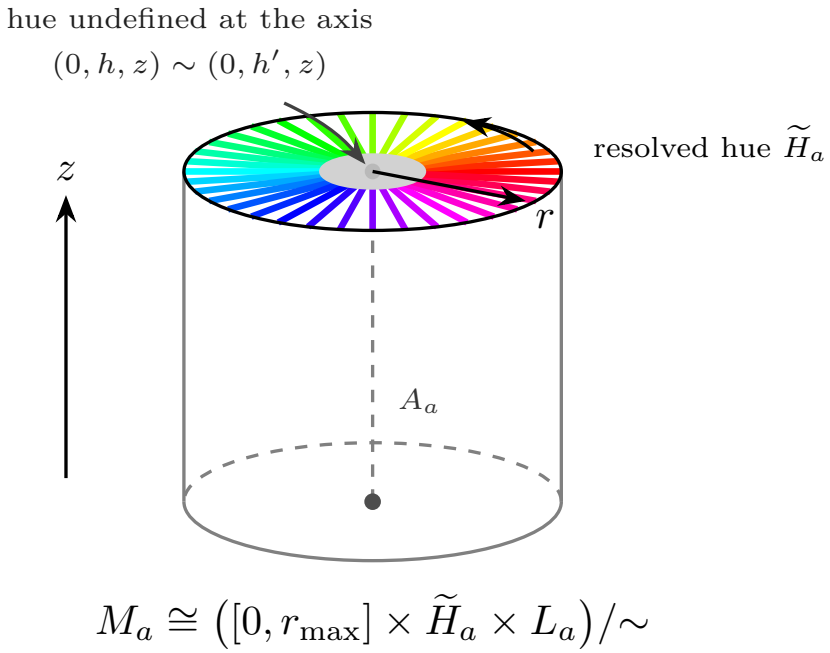


Figure 2: The modeled color body and its achromatic quotient, with a bounded height interval shown. Chromatic radius r is radial, resolved hue $\widetilde{H}_a$ angular, and height z vertical. At $r = 0$, all hue labels are identified at each height, giving the achromatic axis A_a rather than a separate circle of neutral colors. The cylinder represents the chosen ambient target; the ordinary preparation image is a separate subset. Neither the drawn dimensions nor the illustrated colors calibrate perception.

It admits the split-complex notation $q = r + j\kappa z$, $j^2 = 1$, with $q\overline{q} = r^2 - \kappa^2 z^2$. The state-space section is a half-plane because $r \geq 0$; its tangent-space quadratic form has signature $(1, 1)$. This notation introduces no fourth color coordinate. Hue labels the section rather than contributing another component to q.

Hyperbolic rotations preserve the split form on the containing signed plane; they need not preserve the half-plane $r \geq 0$ or the bounded color body. Their use in color theory, including a model of the Hunt effect, has prior precedent [11]. The algebraic invariance alone does not establish a physiological transformation law. The present use of the split form describes the proposed metric and its null directions.

Historical appearance experiments supply separate constraints. Hunt found decreasing saturation as the intensity of the adapting light was lowered [17]. Purdy investigated intensity-dependent changes of spectral hue, associated with the Bezold–Brücke effect [18]. Preserving a split quadratic form alone predicts neither phenomenon. In particular, a transformation confined to one fixed-hue section cannot represent a change of hue; an account of that change must also specify how the hue coordinate depends on intensity and viewing conditions.

The minus sign distinguishes the luminosity direction in the chosen reachability model. A null displacement is not asserted to be a zero perceptual difference. Likewise, a spacelike displacement is not inherently inaccessible: ordinary isoluminant changes have $\mathrm{d}z = 0$ and positive spatial norm. These facts prevent the metric alone from serving as a physiological rule for permitted transitions.

Work on non-Riemannian color differences further illustrates why a local discrimination model and a global appearance-distance model must be distinguished [19]. It does not supply affirmative

evidence for the indefinite metric in Eq. (11). Our metric requires its own justification through the boundary and control assumptions described below.

For a fixed ordinary regime a_0, write

$$C_{a_0} = \operatorname{im} \Phi_{a_0} \subset M_{a_0}. \tag{13}$$

The model assigns the ordinary optical image a spindle-like boundary. To compare it with another regime, both images must first be represented in a common calibrated ambient space. A change of adaptation may then change the response map and its image. The claim that an altered-regime percept lies outside the ordinary image is operational and requires such calibration; the drawing of an exterior region is not itself evidence for that claim.

3.3 Effect geometry and the meaning of height

The symmetric bicone has a standard representation by measurement effects, given in Sec. 5.1 of Berthier [10]. Let

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Q_0 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{14}$$

In normalized coordinates define

$$E(\xi, \eta, \zeta) = \tfrac{1}{2}\left[(1+\zeta)I + \xi R_0 + \eta Q_0\right], \quad r_e = (\xi^2 + \eta^2)^{1/2}. \tag{15}$$

Since $(\xi R_0 + \eta Q_0)^2 = r_e^2 I$, the eigenvalues are $(1 + \zeta \pm r_e)/2$. Hence

$$0 \le E \le I \quad \Longleftrightarrow \quad r_e + |\zeta| \le 1. \tag{16}$$

A unitary change of basis identifies this three-dimensional family with real symmetric two-by-two effects. It is a slice of the four-dimensional complex-qubit effect space. Setting $\xi = x/(\kappa T)$, $\eta = y/(\kappa T)$ and $\zeta = z/T$ gives the symmetric body in Eq. (18). Its two sheets are $\det E = 0$ and $\det(I - E) = 0$.

An effect assigns probabilities $\operatorname{Tr}(\rho E)$ to states ρ; it is not itself a state vector. Identifying effects with perceived colors is an additional model choice. Nor does the matrix representation identify the optical image with the entire effect body. Conjugation by J reverses (ξ, η), complementation $E \mapsto I - E$ reverses all three coordinates, and their composition reverses ζ alone. These symmetries of the ideal body do not imply the same symmetries for a preparation image.

Three height meanings must be distinguished. The opponent interpretation uses a centered white/black activation, with extrema describing focal-versus-surround contrast. In Eq. (15), the height is $\zeta = 2a_3 - 1$, where $a_3 = \operatorname{Tr}(E)/2$ is effect magnitude; its proposed relation to luminance (Sec. 5.3 of Ref. [11]) does not calibrate opponent contrast. In the object-color example of Sec. 6, height is a centered sum of white-normalized cone catches under bounded reflectance. Perfectly absorbing and perfectly reflecting surfaces fix that example's endpoints. Reusing a height coordinate does not identify these quantities.

The ambient cylinder in Eq. (10) and the ordinary image in Eq. (13) are distinct; the model assigns the latter a spindle-like boundary. The cylinder permits chromatic variation at its height extremes. An effect bicone has tips instead. If ordinary black and white are identified with those tips, and a shared cross-regime calibration places an altered chromatic appearance at either of the same extreme heights, Eq. (16) excludes that appearance. This is a conditional incompatibility of those identifications, not a measured location of a chimerical color.

For example, the illustrative point $(r_e, \zeta) = (1/2, -1)$ has eigenvalues $-1/4, 1/4$ and is not an effect. A uniform recalibration

$$E_s = \tfrac{1}{2}\left[(1 + \zeta/s)I + (\xi/s)R_0 + (\eta/s)Q_0\right] \tag{17}$$

with $s > 0$ is positive and bounded by I exactly when $s \geq r_e + |\zeta|$. At $s = 3/2$, the former achromatic references $\zeta = \pm 1$ become $I/6$ and $5I/6$, so they are no longer the tips. Changing the target shape is another possibility. Neither choice is selected by the illustrative point. A full effect target can also contain a proper ordinary image, leaving room for other regimes within the same target. In Secs. 7.3–7.4, Berthier [10] explicitly allows observer-dependent convex subsets of the chromatic state space and leaves contextual effects for further study. We therefore infer no general exclusion of altered colors from the effect approach.

Appendix C states the additional limitation when the two-level coordinate labels paired sectors: on a fixed aligned graph, sector effects have input-independent statistics. Color-sensitive internal observables remain possible, but require a specified encoding and readout.

4 Null-boundary compatibility

4.1 Symmetric normal form

Take $L_{a_0} = [-T, T]$ with $T > 0$. In the isotropic baseline, the prescribed body is

$$C^{\mathrm{sym}} = \{(r, \theta, z) : r \leq \kappa(T - |z|),\ |z| \leq T\}. \tag{18}$$

It is the intersection of the two appropriately oriented cones from the achromatic poles $(0, 0, \pm T)$. For example, a lower-sheet generator is

$$\gamma_\theta(s) = (\kappa s \cos\theta, \kappa s \sin\theta, -T + s), \quad 0 < s < T, \tag{19}$$

and $g_0(\dot\gamma_\theta, \dot\gamma_\theta) = \kappa^2 - \kappa^2 = 0$. The upper sheet has the same property. They meet continuously with different z-slopes at $z = 0$, as illustrated in Fig. 3.

This is a calibration of the metric to the chosen bicone. The linear narrowing toward the poles is not derived from a measurement of gamut shape. Consequently, observing an equatorial ridge would not by itself confirm the metric, and a general empirical color body is not required by topology to possess this ridge. The following narrower statement explains the obstruction within the radial-graph setting.

Proposition 1 (Obstruction to a smooth null radial envelope)**.** *Suppose a compact radial-graph body has boundary $r = R(\theta, z)$ for $-T \leq z \leq T$, with $R(\theta, \pm T) = 0$ and a positive interior value for some fixed θ. Let the metric have no spatial–z cross-terms and a positive-definite spatial block. Then the graph cannot be differentiable everywhere and null everywhere on its positive-radius part.*

Proof. For the stated fixed hue, R has a positive interior maximum in z. If the graph is differentiable there, $R_z = 0$. With $F = r - R$, the conormal is $\mathrm{d}F = (1, -R_\theta, 0)$. Because the inverse spatial block is positive definite, $g^{-1}(\mathrm{d}F, \mathrm{d}F) > 0$, contradicting nullity. □

The proposition uses the radial-graph and metric hypotheses. It forces a failure of differentiability somewhere if the entire boundary is to remain null; it does not locate that failure at an isoluminant equator for an arbitrary body. Eq. (18) supplies that location through symmetry. The achromatic tips are separate singular points of the boundary graph.

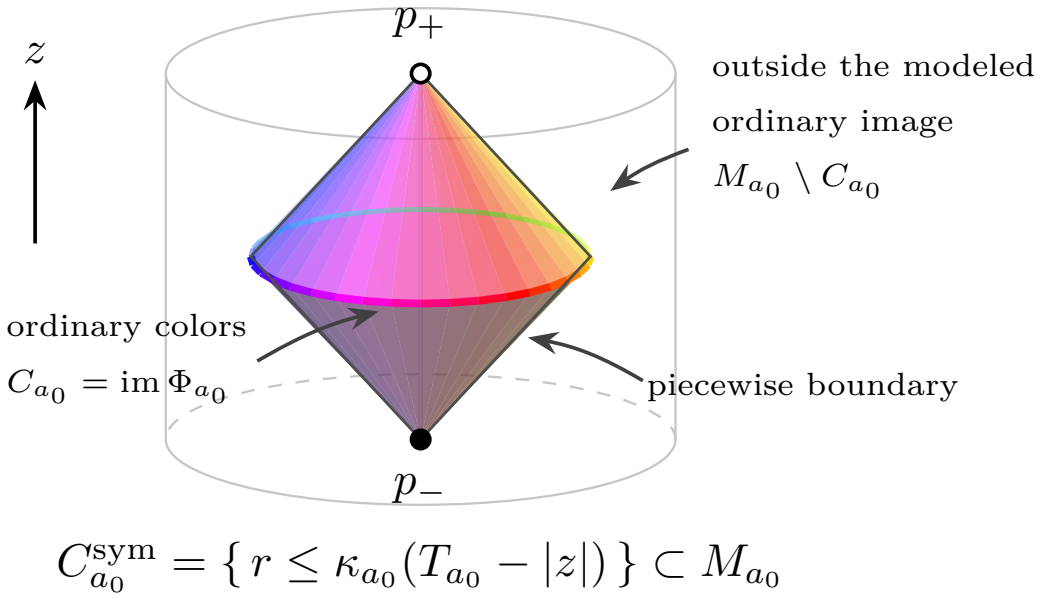


Figure 3: Prescribed symmetric reachability body inside the ambient color model. The open upper and lower sheets are null for the calibrated baseline metric and meet at an equatorial ridge. The equatorial location follows the assumed symmetry. Exterior points are candidates for comparison across viewing regimes; their physiological accessibility is not implied by the metric.

4.2 Hue-dependent ceiling and general spatial block

Let $\alpha(\theta) > 0$ be smooth and periodic, and prescribe Eq. (1). On either open sheet, away from the ridge and tips, consider

$$\mathrm{d}s^2 = A\,\mathrm{d}r^2 + 2D\,\mathrm{d}r\,\mathrm{d}\theta + B\,\mathrm{d}\theta^2 - C\,\mathrm{d}z^2, \tag{20}$$

where $A > 0$, $C > 0$, and $\Delta = AB - D^2 > 0$. These conditions make the spatial block positive definite and give signature $(2, 1)$. The absence of spatial–z cross-terms is part of the ansatz. All coefficients below are evaluated on the boundary unless stated otherwise.

For $F = r - R(\theta, z)$, the null-conormal equation is

$$\frac{B + 2DR_\theta + AR_\theta^2}{\Delta} - \frac{R_z^2}{C} = 0. \tag{21}$$

The original model fixes the luminosity coefficient on the boundary by

$$C = \alpha(\theta)^2. \tag{22}$$

Since $R_z = \mp\alpha$ on the two sheets, $R_z^2/C = 1$. This calibration is essential. Without it, the null condition relates the same spatial expression to a different luminosity factor and need not yield the bound below.

Multiplying Eq. (21) by Δ gives

$$B + 2DR_\theta + AR_\theta^2 = \Delta. \tag{23}$$

Multiplication by B, followed by elimination of $AB = \Delta + D^2$, yields

$$\boxed{\Delta(B - R_\theta^2) = (B + DR_\theta)^2.} \tag{24}$$

Thus every regular spatial block satisfying the calibrated null condition obeys $R_\theta^2 \le B$. Defining

$$\beta(\theta, z) = \frac{\sqrt{B(R(\theta, z), \theta, z)}}{R(\theta, z)}, \tag{25}$$

and using $R_\theta = (\alpha'/\alpha)R$ proves Eq. (2). The result is a necessary compatibility condition for the specified boundary, calibration, and spatial metric. It is not a restriction imposed by signature alone on every possible choice of color coordinates and metric.

The independence from the displayed coefficient D does not make the result invariant under arbitrary changes mixing r, θ, and z. Such changes may alter the boundary graph, the absence of temporal cross-terms, or the calibration. For a pure hue change $\psi = \psi(\theta)$, however,

$$\left|\frac{\mathrm{d}\log\alpha}{\mathrm{d}\psi}\right| = \left|\frac{\mathrm{d}\theta}{\mathrm{d}\psi}\right| \left|\frac{\mathrm{d}\log\alpha}{\mathrm{d}\theta}\right|, \qquad \beta_\psi = \left|\frac{\mathrm{d}\theta}{\mathrm{d}\psi}\right| \beta_\theta. \tag{26}$$

Both sides transform identically. This is the precise reparameterization covariance asserted here.

4.3 Regular diagonal construction

Theorem 2 (Diagonal boundary compatibility)**.** *In the radial–angular diagonal ansatz $D = 0$, with $B > 0$ and the calibration of Eq.* (22)*, a finite positive boundary value of A satisfying nullity exists if and only if $R_\theta^2 < B$. It is uniquely given by*

$$A = \frac{B}{B - R_\theta^2} = \left[1 - \left(\frac{\alpha'}{\alpha\beta}\right)^2\right]^{-1}. \tag{27}$$

Proof. Eq. (23) with $D = 0$ is $B + AR_\theta^2 = AB$. Solving gives Eq. (27). Its denominator is positive precisely under the strict inequality. Equality gives no finite solution; a negative denominator gives $A < 0$. □

The diagonal condition is a restriction of the model, not a harmless coordinate simplification that necessarily preserves the specified R, B, and C. In the general spatial block, equality in the non-strict bound can be regular. At a point, take

$$R_\theta = 1, \quad B = 1, \quad D = -1, \quad A = 2. \tag{28}$$

Then $\Delta = 1$, Eq. (23) holds, and all coefficients are finite, despite $R_\theta^2 = B$. The divergence in Eq. (27) therefore describes failure of a finite diagonal realization. It does not establish a universal geometric horizon or a removable coordinate singularity.

On a compact patch of an open sheet where the strict inequality has a positive margin, the diagonal boundary data extend smoothly to a collar and remain positive by continuity. They may be interpolated toward a Cartesian core using a cutoff that vanishes on a neighborhood of the interior achromatic axis. Positivity of the spatial block is preserved by convex interpolation of positive-definite spatial tensors, while the temporal coefficient remains negative. The construction is local away from the ridge and achromatic boundary tips; Appendix A records its scope.

4.4 Constraint and optical non-egress

For the ordinary image to be a forward-invariant reachable set, an additional statement about response dynamics is required. Let $\mathcal{V}_a(x)$ denote the physically admissible one-sided control velocities at a boundary state. A necessary boundary condition is

$$v(F) \le 0 \qquad (v \in \mathcal{V}_a(x),\ F(x) = 0), \tag{29}$$

on each smooth sheet, with corresponding conditions for every active boundary constraint at a ridge. Suitable regularity and evolution hypotheses would be needed to turn this tangency condition into a full invariance theorem.

Raising the conormal with the metric does not derive Eq. (29). If $n^\mu = g^{\mu\rho}\partial_\rho F$, then for every vector v,

$$g_{\mu\nu} n^\mu v^\nu = g_{\mu\nu} g^{\mu\rho} \partial_\rho F \, v^\nu = v(F). \tag{30}$$

The metric cancels. The condition concerns the controls and the boundary, independently of signature. Moreover, $\operatorname{im} \mathrm{d}\Phi_a$ is a linear space when Φ_a is differentiated on an unconstrained smooth domain: if it contains v, it contains $-v$. Requiring non-egress for both would force $v(F) = 0$. A useful one-sided condition must therefore concern the admissible controls, including any saturation constraints, rather than the entire unconstrained differential image throughout a collar.

Opponent-channel saturation is a possible physiological source of such control restrictions. It is not established here. Since ordinary isoluminant optical changes are spacelike in the model, nullity alone supplies no causal confinement argument. The established result is the compatibility constraint in Eq. (24); physiological non-egress remains a separate hypothesis.

5 Numerical model and calibration

5.1 Cone coordinates and the meaning of the test

For a computational illustration, we use the packaged Stockman–Sharpe 2° cone fundamentals in Colour Science 0.4.7, whose underlying cone-sensitivity studies include Refs. [20, 21]. The input arrays contain 441 wavelength samples from 390 to 830 nm. The stimulus model uses nonnegative mixtures of the 311 tabulated response vectors from 390 to 700 nm. Its normalized attainable set is the convex hull of their chromaticities: normalizing a positive sum of response vectors gives a convex combination weighted by each component's total response, and conversely every such combination is attainable.

The calculation therefore uses this convex hull as its outer mixture boundary. It does not assume that every point of the spectral locus lies on that boundary. The same hull is obtained if response vectors between the retained wavelengths are interpolated linearly. This is an exact statement about the declared discretized stimulus model, not about the entire continuous human spectral response or an optimized passive object-color solid.

Let E_i be the integrated response of cone i to equal-energy illumination over the tabulated interval. In the primary convention we first define

$$\widetilde{L} = \frac{L}{E_L}, \qquad \widetilde{M} = \frac{M}{E_M}, \qquad \widetilde{S} = \frac{S}{E_S}, \tag{31}$$

and then normalize their sum to obtain (l, m, s). Equal-energy white is now $(1/3, 1/3, 1/3)$. In the raw-fundamental comparison, the cone functions are left unchanged and white is their actual integrated chromaticity. Centering raw fundamentals on the barycenter without first rescaling them would combine two different conventions.

In either convention define $x = l - l_w$, $y = m - m_w$, $r = (x^2 + y^2)^{1/2}$, and $\theta = \operatorname{atan2}(y, x)$. Set $T = 1$ for this illustration, so the equatorial radial profile is $\alpha(\theta)$; for another T, it is R_{eq}/T and its logarithmic angular derivative is unchanged. This is a colorimetric radius in the declared chart, not a direct measurement of perceived saturation. Its use in Eq. (1) supplies the equatorial profile of the assumed separable boundary; the calculation does not test its full dependence on luminosity.

The simplex Fisher metric [22, 23] has

$$G = \begin{pmatrix} l^{-1} + s^{-1} & s^{-1} \\ s^{-1} & m^{-1} + s^{-1} \end{pmatrix}, \qquad s = 1 - l - m. \tag{32}$$

It is positive definite for $l, m, s > 0$. For the fixed-radius angular direction, $\partial_\theta(l, m) = r(-\sin\theta, \cos\theta)$, so its angular coefficient is

$$B_{\mathrm{disc}} = r^2\big(G_{ll}\sin^2\theta + G_{mm}\cos^2\theta - 2G_{lm}\sin\theta\cos\theta\big). \tag{33}$$

The modeled identification $B = B_{\mathrm{disc}}$ then supplies $\beta = \sqrt{B_{\mathrm{disc}}}/r$. This uses the angular component at fixed radius. It is not the metric pullback along the gamut curve, whose tangent also contains $\alpha'(\theta)$.

Normalized cone coordinates alone do not establish a multinomial perceptual noise law. Eq. (32) is a tractable information-geometric choice, not an independently calibrated discrimination tensor. Its statistical interpretation and overall scale would require a measurement model, specified luminance, and an integration window. The Fisher form is imposed separately in the raw and balanced conventions, rather than transporting one fixed tensor between them. The comparison therefore changes the metric identification as well as the coordinates; it is not a test of the coordinate covariance of one unchanged tensor.

5.2 Recalculation and numerical scope

The convex hull has 188 vertices in both conventions. Its short-to-long mixture edge joins the retained 410 and 700 nm vertices; additional edges bridge smaller nonconvex portions of the tabulated locus. The earlier 390–700 nm endpoint chord and the entire monochromatic arc therefore do not describe the outer attainable boundary. In particular, the near-403 nm feature discussed below is interior to the mixture hull.

For a hull edge parameterized by $p(t) = p_0 + tv$, where $p = (x, y)$ is centered on the chosen white, direct differentiation gives

$$\frac{\mathrm{d}\log r}{\mathrm{d}\theta} = \frac{p \cdot v}{p_x v_y - p_y v_x}. \tag{34}$$

This evaluates the angular slope within each open edge without finite-difference averaging across corners. We sample 101 interior points per edge, giving 18,988 points in each convention. At vertices, the ordinary derivative is not defined; the edge formulas describe the two one-sided limits. The finite sampling is not an interval certificate for all intervening points and does not assert a globally smooth ceiling for the polygon.

The simplex metric is undefined when a cone coordinate is zero and grows large near such zeros. For continuity with the manuscript's diagnostic convention, the primary summaries retain finite points with $\beta < 10$, evaluated at metric scale $c = 1$. This is a computational filter, not a threshold defining physiological nondegeneracy. The supplied table has zero short-cone sensitivity from 616 nm onward, making explicit exclusion of singular values essential.

Every retained edge sample satisfies the strict diagonal condition at the chosen scale (Table 1). Figure 4 shows the corresponding angular profiles. These results are evidence of compatibility for the sampled model, not a fitted or independently verified law of human discrimination.

Table 1: Compatibility on sampled open hull edges

Quantity	E-balanced	Raw, E-centered
Retained samples	14,436	12,267
Samples with $u < \beta$	14,436	12,267
Minimum $\beta - u$	1.0085	0.8821
Maximum u/β	0.6133	0.6347
Sampled c_*	0.3761	0.4028

$u = |\alpha'/\alpha|$; 101 samples per open edge of the 188-vertex hull. Retention uses finite $\beta(c = 1) < 10$. The sampled c_* is defined in Eq. (35). Neither the grid nor the filter establishes an all-hue physiological bound.

A control calculation explains why the boundary definition and differentiation method matter. On the original spectral-locus construction, a 1° angular grid misses a narrow raw-normalization failure near 403 nm. Differentiating a PCHIP interpolation of the LMS functions resolves it: at 403.4 nm, $u \approx 5.9852$ and $\beta \approx 5.2312$, so $(u/\beta)^2 \approx 1.3091$. This feature is inside the attainable mixture hull and is therefore not a failure of the outer-boundary check in Table 1. It is retained in the reproducibility archive to document why the coarse original-locus result cannot support an all-hue claim.

5.3 Metric scale and a testable comparison

Writing $u = |\alpha'/\alpha|$, rescaling the chromatic metric $G \mapsto cG$ while holding $C = \alpha^2$ fixed gives $\beta \mapsto \sqrt{c}\,\beta$. On a fixed retained finite grid, strict compatibility is equivalent to

$$c > c_* := \max_\theta \left(\frac{u(\theta)}{\beta(\theta)} \right)^2 . \tag{35}$$

The value $c = 1$ is a convention in this illustration, not a measured physiological calibration. Increasing it can turn a failure into a pass without changing the boundary. Thus the test is hue-coordinate covariant but not independent of discrimination-metric normalization. The two properties must not be conflated.

A common rescaling of the entire Lorentzian metric leaves nullity unchanged. Photon number alone therefore does not calibrate the relative chromatic and height scales. For example, independent Poisson catches with means Np_i, $\sum_i p_i = 1$, have Fisher form

$$\sum_i \frac{\mathrm{d}(Np_i)^2}{Np_i} = N \left(\sum_i \frac{\mathrm{d}p_i^2}{p_i} + \mathrm{d}\zeta_I^2 \right), \qquad \zeta_I = \log N. \tag{36}$$

The cross-term vanishes because $\sum_i \mathrm{d}p_i = 0$. Both parts scale with N; the negative sign of a reachability metric is an additional choice, not a property of Fisher information. This calculation supplies no lower bound on a working observer's relative discrimination scale.

For the Euclidean spatial reference metric, $B = r^2$ and $\beta = 1$ in the stated polar coordinate. Comparing u with 1 is a comparison with that specific metric. If the angle is reparameterized and the Euclidean tensor is transformed consistently, its comparison is covariant too. A failure of the fixed Euclidean benchmark does not, by itself, demonstrate a coordinate artifact or select the Fisher metric as physiological truth.

An empirical test would measure the ceiling and local discrimination independently in the same observer, stimulus class, adaptation state, coordinates, and units. It would propagate their uncertainties and fix the metric scale before evaluating the inequality. A large apparent margin

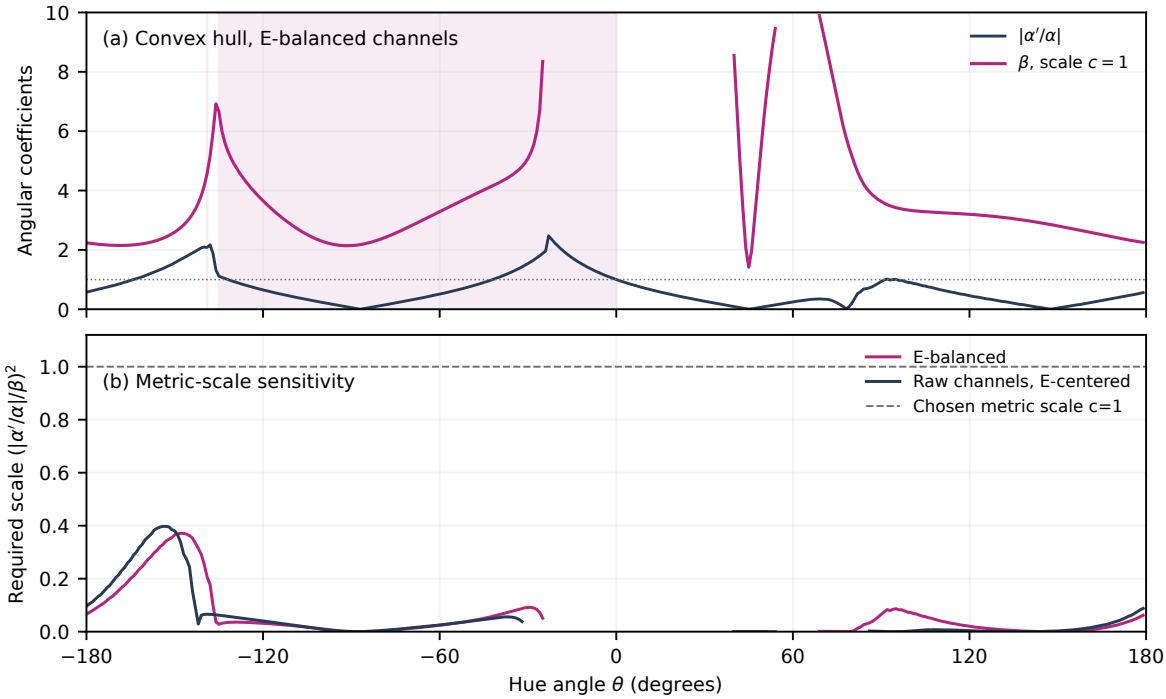


Figure 4: Numerical compatibility for the convex hull of the selected 390–700 nm response vectors. (a) E-balanced profiles of $u = |\alpha'/\alpha|$ and the fixed-radius Fisher scale β; the dotted line marks the Euclidean reference $\beta = 1$. Pink shading marks hull edges joining nonadjacent sampled wavelengths, not a perceptual hue class. (b) Required metric scales $(u/\beta)^2$ in both normalizations. The curves use a 1° grid; Table 1 uses denser open-edge samples. Values with $\beta \geq 10$ are omitted from the β trace and panel (b). The filter and metric scale are modeling choices. The vector figure was generated with Matplotlib 3.10.8 by Python code developed with OpenAI ChatGPT/Codex assistance; it contains calculated model quantities, not experimental measurements.

obtained by comparing differently normalized metrics cannot supply that identification. Comparisons with MacAdam ellipses [24] require a common metric scale and a justified relation between interior measurement locations and the boundary. Without those identifications, they do not independently validate the boundary inequality.

One possible calibration identifies $B = kB_{\text{meas}}$ and $C = kC_{\text{meas}}$ with chromatic and height sensitivities in the same units, with one constant $k > 0$. The condition $C = \alpha^2$ then requires C_{meas}/α^2 to be constant on the tested boundary; it is itself a prediction of this identification. Once that ratio is fixed, the angular constraint becomes

$$R_\theta^2\, C_{\text{meas}} \leq \alpha^2 B_{\text{meas}}, \tag{37}$$

with strict inequality for the diagonal realization. This is an additional measurement bridge, not a consequence of cone fundamentals. It leaves the radial and radial–angular coefficients free. Identifying the entire spatial block with a scaled Fisher tensor is a stronger choice, excluded on nontrivial open edge segments where that tensor is defined by Appendix B.

6 A bounded-reflectance obstruction

6.1 Timelike directions and boundary conormals

The compatibility theorem concerns a prescribed bicone. A different question is whether an independently specified signal body can have an everywhere-null boundary under a chosen interpretation of height. The following local fact allows all metric cross-terms.

Proposition 3 (Achromatic tangency obstruction)**.** *Let a boundary in a three-dimensional signal space be differentiable at a point, with nonzero conormal n. If $n(W) = 0$ for a direction W, then every Lorentzian metric of signature $(2, 1)$ with $g(W, W) < 0$ at that point satisfies*

$$g^{-1}(n, n) > 0. \tag{38}$$

In particular, the boundary is not null there.

Proof. Put $n^\sharp = g^{-1}n$. Then $g(n^\sharp, W) = n(W) = 0$. The orthogonal complement of a timelike vector is positive definite. Since $n^\sharp \neq 0$, its squared norm is strictly positive. □

The hypothesis concerns a tangent direction, not a coordinate name. Transporting the metric, conormal and direction together under a coordinate change preserves the result. For a differentiable radial graph at an interior maximum in height, the tangent plane contains ∂_z, so this also strengthens the metric scope of Proposition 1 whenever that direction is timelike.

6.2 Continuum signal model and a unique normal

We now use a different stimulus class from the additive mixture hull in Sec. 5. Let $s_i(\lambda)$ be the componentwise PCHIP interpolants of all 441 Stockman–Sharpe samples on $[390, 830]$ nm. Set

$$\begin{aligned} c_i(\lambda) &= \frac{s_i(\lambda)}{\int_{390}^{830} s_i(\lambda)\,\mathrm{d}\lambda}, \\ \mathcal{O} &= \left\{ \int_{390}^{830} c(\lambda)\rho(\lambda)\,\mathrm{d}\lambda : 0 \le \rho \le 1 \right\}. \end{aligned} \tag{39}$$

Here ρ is any measurable reflectance, and equal-energy illumination is included in c. White is exactly $W = (1, 1, 1)$ in catch coordinates. This is the standard bounded-reflectance construction [13]; it is not the cone of unrestricted additive spectra. Its support function is

$$h(n) = \int_{390}^{830} \max\{n \cdot c(\lambda), 0\}\,\mathrm{d}\lambda. \tag{40}$$

It is attained by choosing $\rho = 1$ where $n \cdot c > 0$ and $\rho = 0$ where $n \cdot c < 0$. This pointwise rule is independent of any perceptual metric.

Replacing ρ by $1 - \rho$ sends a to $W - a$, so $\mathcal{O}$ is centrally symmetric. In the chart introduced below in Eq. (42), this is $(x, y, z) \mapsto (-x, -y, -z)$; its radial boundary therefore obeys

$$R_{\mathcal{O}}(\theta, z) = R_{\mathcal{O}}(\theta + \pi, -z). \tag{41}$$

The upper polar tangent cone is the reflection of the lower. A boundary with one ceiling α on both sheets, as in Eq. (1), can match both polar tangent cones in this chart only if $\alpha(\theta + \pi) = \alpha(\theta)$. This necessary condition does not establish the separable bicone form. Central symmetry alone does not imply height reflection at fixed hue.

Proposition 4 (Two independent transition vectors)**.** *Suppose $\mathcal{O}$ is full dimensional and c is continuous. A genuine interior $0 \leftrightarrow 1$ transition means that ρ equals zero almost everywhere on one adjacent interval and one almost everywhere on the other. If a boundary point is represented by a reflectance with at least two such transitions, and two of the corresponding response vectors are linearly independent, the point has a unique outward normal direction and the boundary is differentiable there.*

Proof. Any supporting functional at this point is maximized by the same reflectance. The nonnegative difference between its pointwise maximum and its attained integral must vanish. Continuity therefore forces the functional to vanish at each genuine transition. Two independent transition vectors determine a one-dimensional space of possible normals. Full dimensionality excludes both orientations supporting the same point, leaving one outward normal ray. For a convex body this is differentiability at the point. □

Additional transitions do not invalidate the argument. Merely touching zero without a reflectance transition is insufficient. No continuously differentiable neighborhood is asserted.

6.3 Certified instance and its scope

For a catch vector a, choose the affine coordinates

$$\begin{aligned} x &= \tfrac{2}{9}(2a_L - a_M - a_S), \\ y &= \tfrac{2}{9}(-a_L + 2a_M - a_S), \\ z &= \tfrac{2}{3}(a_L + a_M + a_S) - 1. \end{aligned} \tag{42}$$

Black and white lie at $(0, 0, -1)$ and $(0, 0, 1)$. The *direction* W is sent by the linear part to $(0, 0, 2)$. Take the support conormal $n = \mathrm{d}y$, or, equivalently, maximize the integral of

$$q(\lambda) = -c_L(\lambda) + 2c_M(\lambda) - c_S(\lambda). \tag{43}$$

The certificate treats the archived binary64 samples as exact rational data and constructs the PCHIP coefficients with rational arithmetic. Signs on the interpolation cells are certified by Bernstein-polynomial bounds, with subdivision where required; the two transition cells have derivative signs of fixed nonzero sign and their roots are enclosed by rational bisection. The cost has exactly two simple roots, with $q > 0$ between them and $q < 0$ outside. Outward-rounded enclosures are

$$\begin{aligned} &486.8630809758 < \lambda_1 < 486.8630809761, \\ &599.3694477603 < \lambda_2 < 599.3694477606 \end{aligned} \quad \text{nm}. \tag{44}$$

Thus $\rho_* = \mathbf{1}_{(\lambda_1, \lambda_2)}$ is a support optimizer. Exact polynomial integration and outward interval bounds enclose its point by

$$\begin{aligned} 0.11649647725622 \le x_* \le 0.11649647725683, \\ 0.20194541250715 \le y_* \le 0.20194541250768, \\ 0.09328313559491 \le z_* \le 0.09328313559570. \end{aligned} \tag{45}$$

At either root, $2c_M = c_L + c_S$, so the transformed response vector is

$$((c_L - c_S)/3,\ 0,\ c_L + c_S). \tag{46}$$

The interval calculation gives $-0.003 < c_L - c_S < -0.002$ at the first root and $0.007 < c_L - c_S < 0.008$ at the second, with $c_L + c_S > 0$ at both. The vectors are independent. A response vector inside the

positive band has positive y component; together these span three dimensions, so by continuity $\mathcal{O}$ is full dimensional. Proposition 4 therefore gives a unique tangent plane at a_*, with conormal $\mathrm{d}y$. Since $\mathrm{d}y(W) = 0$, Proposition 3 excludes nullity for every Lorentzian metric making the adding-white direction timelike there.

The same signs show that a positive combination of the two transition vectors points along W: their chromatic components cancel. These are complementary spectral directions in the specified model. Tangent planes parallel to the achromatic axis and complementary transitions belong to the classical geometry of full colors (Koenderink [12], p. 213). The calculation makes no claim that this geometry is new and imposes no universal two-transition restriction on object-color boundaries.

The certificate concerns a continuous piecewise-cubic response density and all measurable reflectances in $[0, 1]$, not a finite-bin reflectance polytope. It certifies the declared interpolation model, not biological sensitivities without uncertainty. No sampled proportion of the hue circle is needed. The point is above the equator, but its height does not drive the obstruction: achromatic tangency does.

This result prevents identifying the entire boundary of $\mathcal{O}$ with an everywhere-null surface while retaining the stated timelike direction. It does not refute the conditional bicone theorem or determine a perceptual boundary. A merely different coordinate chart cannot remove the obstruction. A different response encoding must explain what happens to the boundary and the achromatic direction; nonlinearity alone does not guarantee an escape.

7 Discussion

7.1 The transformation problem

The motivating question is how stimulus conditions, observer state, and context are related to the organization of color as it appears. A mechanistic account addresses this question by relating its physical variables to specified appearance judgments and their regularities. A geometric account addresses it by stating those relations in a target space and testing the map into it. These approaches can inform one another: the geometry expresses relationships that a realization can explain or fail to reproduce, while physiology constrains the maps and controls that the geometry may represent.

The compactification construction provides one description of spectral and nonspectral arcs within a cyclic hue target. It does not require the visual system to glue wavelengths or implement the construction as a neural operation. Nor does the comparison between a wavelength interval and a hue circle establish a failure of physical explanation. The actual domain contains mixtures, and the trichromatic signal gamut already has a closed boundary. The remaining identification is between signal relations and appearance relations, including which experimental procedures establish them. A color-matching quotient, a hue ordering, and a discrimination metric answer different questions. In particular, the equivalence in Eq. (5) is relative to the selected response map and fixed regime; it does not assert equality of every appearance attribute in every context.

This also clarifies what is observer-relative in the construction. Spectral sensitivities and viewing conditions enter the response map; the selected spectral arc, its limiting response classes, and the calibration of the hue circle must be specified for that map. The topological construction alone predicts neither another observer's spectral limits nor a uniform spacing of hues. Its contribution is to make the spectral/nonspectral distinction compatible with a common cyclic target while leaving those empirical identifications explicit.

7.2 Context, presentation, and accessible appearances

Blackness induction gives a concrete example of why the focal stimulus is not a complete appearance specification. Werner and colleagues held a central field fixed while increasing an annular surround until the center appeared black [25]. Volbrecht and colleagues varied a preceding inducing field and measured changes in the blackness of a fixed reference stimulus [26]. Spatial context and temporal history therefore belong in the response conditions. These experiments establish changes in appearance under specified manipulations; they do not establish the full set of appearances attainable in the absence of contrast, or prove that one regime's entire image strictly contains another's.

Changing a response map and enlarging its image are separate claims. Even when a fixed stimulus changes appearance, the two maps may have the same range. A reachability comparison requires a common operational identification of appearances across regimes and evidence that a particular appearance is accessible in one regime and unavailable in the other. A strict inclusion of whole images is stronger still. The ambient cylinder of Eq. (10) and the spindle assigned to the ordinary image provide a model in which to formulate such questions; the unused part of a drawing does not establish perceptual possibilities or their accessibility.

Chimerical-color descriptions, including stygian blue and self-luminous green, motivate examining combinations of chromatic and white/black appearance under altered adaptation [27]. They do not, without a shared calibration, locate an appearance at an extreme height of the present model or outside the ordinary image. Reports of reddish green and yellowish blue under special viewing conditions concern a related but distinct question [28]. Afterimages also require care: their occurrence after removal of an inducing stimulus does not imply that their appearance has no ordinary physical match. Matching and nulling procedures have compared afterimages with physically presented stimuli [29]. These distinctions allow regime changes to be studied without making every unusual appearance evidence for an enlarged color space.

The present target is color as visually presented under the conditions being compared. Naming a color, recalling it, imagining it, and seeing it need not be modeled by one undifferentiated report procedure. This paper leaves a general account of modes and degrees of presentation open; it does not classify all memory or imagery as non-vivid, or infer a separate state space from vividness alone. An afterimage illustrates presentation without the corresponding inducing stimulus at that moment, not a universal distinction between perception and memory.

7.3 Signal limits and appearance criteria

The three meanings of height in Sec. 3.3 matter in comparisons with experiments. White-normalized catches in Eq. (39) use a fixed perfect-reflector reference under a fixed illuminant. That normalization does not by itself model the surround as the observer's reference white. With focal catches and that reference fixed, changing the surround does not move the signal point, even when it changes appearance. An encoding that instead estimates its white reference from the scene is a further model, whose coordinates and predictions must be specified.

Luminosity-threshold studies offer a more direct comparison between signal limits and an appearance criterion. Morimoto and colleagues compared settings for the transition between surface and self-luminous appearance with the largest luminance attainable by a bounded reflectance at each tested chromaticity [30]. For the specified illuminant and response model, each such upper limit is a boundary point of the associated bounded-reflectance solid, reached along a fixed-chromaticity ray from black. The reported relationship concerns the shape of the threshold locus and its dependence on illumination; it does not identify every threshold with the physical limit. Duay and

Nagai extended this comparison to targets in natural-scene images [31]. Some observers judged implausible brightness when the targets did not appear self-luminous, so the report criterion must also be retained. These studies motivate testing a signal-to-report relation. They do not establish that the particular boundary certified here is a perceptual boundary or is null for a perceptual metric. Luminous appearance is itself ordinary: a signal beyond a passive-reflectance gamut need not lie beyond the stimuli available from light sources, much less beyond all ordinarily accessible appearances.

The two signal calculations in this paper also have different domains. Section 5 uses an additive mixture hull and a modeled angular information metric. Section 6 permits all bounded reflectances against a specified continuous response density. Neither measures an individual's appearance boundary or discrimination tensor. The physical stimulus class must remain part of any comparison with reported appearance; a boundary for one class cannot silently serve as a boundary for every class.

7.4 What the geometry contributes

The metric calculation gives a conditional result with a definite mathematical content. Once the boundary form, the lack of temporal cross-terms, the calibration $C = \alpha^2$, and the angular coefficient B have been prescribed, Eq. (24) supplies a necessary inequality. In the diagonal ansatz, strict inequality is sufficient for finite positive boundary data and a local regular collar. The construction neither singles out a unique interior metric nor selects the angular metric from cone fundamentals. A successful numerical illustration therefore establishes consistency under its conventions, rather than a calibrated account of human discrimination.

There are distinct aspects of the ridge claim. Within the symmetric bicone, the ridge at $z = 0$ is an exact feature of the assumed boundary. Proposition 1 gives a broader obstruction to a differentiable, everywhere-null radial envelope. Proposition 3 permits cross-terms and applies to the certified point of the separately specified signal solid. Thus an everywhere-null boundary with adding white timelike is already excluded for that model. The certificate establishes a unique tangent plane at the stated point, not a smooth neighborhood or a percentage of the boundary. None of these results shows that every perceptual color solid has an isoluminant ridge.

The effect representation adds an algebraic comparison. An ordinary preparation image can be a proper convex subset of a carrier while spanning it linearly. Its excluded colors then describe a convex reachability restriction, not an adjoint-kernel defect. For example, a linear or affine encoding that keeps three collinear signal points distinct cannot send all three to distinct pure chromatic states on a round disk: a line intersects the circle in at most two points. This constraint applies only to encodings with those assumptions. The tabulated long-wavelength chromaticities with $S = 0$ supply collinear examples, but the exact zeros belong to the declared input table. Appendix C further shows how internal color-sensitive observables may be paired between carriers without identifying hue with an intersector phase or assigning inaccessible colors to a linear complement.

A claim of confinement requires additional control data. To test non-egress, an experiment or physiological model would need to identify admissible controls near the candidate boundary and determine whether outward velocities remain available. Eq. (30) makes clear why invoking a null normal cannot substitute for that task. A successful control model could support a reachability interpretation without making ordinary optical motion globally causal in the Lorentzian sense.

7.5 Realization and philosophical scope

The next realization must specify a stimulus and adaptation encoding, an observer-state carrier, and observables or report procedures that define the color coordinates. It must then distinguish the predicted appearance image from the larger target in which that image is represented. Existing measurements can constrain this construction when their stimulus classes, viewing regimes, coordinates, and report criteria match it. Independent boundary and discrimination measurements would test Eq. (2) only after the metric identification and relative scale are fixed. Control measurements address confinement separately. The resulting constraints can rule out a proposed realization even though the underlying geometric construction is mathematically consistent.

The transformation problem can be pursued without settling the metaphysics of experience. Here "appearance space" names the organized target of the psychophysical account; it does not establish a particular ontology of experience. A structural account would explain relationships among preparations, observer conditions, and appearances. Whether such an account exhausts what needs explaining about experience belongs to the further debate over consciousness and physicalism [32, 33]; the present construction does not settle that debate. The present contribution is to formulate candidate relationships precisely enough that compatibility, failure, and missing empirical identifications can be distinguished.

8 Conclusion

The magenta gap motivates a response-side description of hue closure that distinguishes physical spectral stimuli from the cyclic organization of color. Inserting a purple interval into an auxiliary compactified spectral arc gives a resolved hue circle; collapsing that circle at zero chromatic radius gives a color body with an achromatic axis. These constructions provide the setting for a proposed reachability metric, without determining its signature.

For a prescribed hue-dependent bicone and explicit metric calibration, nullity and positive spatial definiteness imply a coordinate-covariant bound on the angular variation of the ceiling. The diagonal realization requires strict inequality, while regular equality can occur with a radial–angular cross-term. Cone-fundamental calculations illustrate compatibility with a specified metric convention. Independent discrimination calibration and a model of admissible optical controls remain necessary to establish empirical boundary constraints and physiological confinement.

The same symmetric body admits an established two-level effect representation, but effect magnitude, opponent contrast and bounded-reflectance signal height are distinct quantities. The certified object-color point supplies a definite limit: its boundary cannot be null for any Lorentzian metric that makes adding white timelike there. This constrains a proposed signal-to-geometry identification without deciding the geometry of perception. A stimulus encoding and an operational readout remain necessary to turn either representation into a perceptual model.

A Scope of the regular extension

The local extension used after Theorem 2 is a statement about compact patches of an open boundary sheet. Extend the smooth positive boundary coefficients along a transverse collar coordinate. A positive lower bound for $1 - u^2/\beta^2$ on the patch keeps A finite. A partition of unity can join its spatial tensor to a regular Cartesian spatial tensor away from the boundary, because positive-definite matrices form a convex cone. Interpolate the positive coefficient C separately. Cutoffs are chosen to vanish on a full neighborhood of the interior axis, so arbitrary angular dependence does not survive

at $x = y = 0$.

This is stronger than requiring a correction merely to be $O(r^2)$: $r^2 f(\theta)$ need not be smooth in Cartesian coordinates for arbitrary f. It also avoids extending a formula involving $|z|$ through the equator as though that formula were smooth. Separate sheet collars are sufficient for the local boundary statement; a globally smooth realization across all other regions is an additional construction problem.

There is a direct obstruction at the achromatic boundary tips for nonconstant α. Along the boundary, $C = \alpha(\theta)^2$. Approaching $(0, \pm T)$ at different hues gives different limits for the coefficient of $\mathrm{d}z^2$. The stipulated calibration therefore does not generally extend continuously to those tips. The regular Cartesian core concerns the open interior axis, not the closure including the anisotropic achromatic poles. No global smoothness or removable-singularity claim at those poles is required for the local compatibility theorem.

B Why the full Fisher identification fails on an edge

The angular identification leaves A and D available for the construction. Suppose instead that the whole spatial block is cG, with a single constant $c > 0$ and G the simplex Fisher tensor. At a fixed height on an open boundary sheet, let a nontrivial straight edge segment lie in the simplex interior $l, m, 1 - l - m > 0$. Let its outward normal be ν and its equation $\nu \cdot (p - w) = d > 0$, where $p = (l, m)$ and w is white. Put $e_r = (p - w)/r$. The spatial conormal of $F = r - R$ is $\nu/(\nu \cdot e_r)$, since its radial component is one. The calibrated null condition therefore requires

$$c = \frac{\nu^{\mathsf{T}} G^{-1} \nu}{(\nu \cdot e_r)^2} = \frac{r^2 \nu^{\mathsf{T}} \Sigma(p) \nu}{d^2}, \qquad \Sigma(p) = \mathrm{diag}(l, m) - pp^{\mathsf{T}}. \tag{47}$$

Here p, r, d are evaluated on the chosen height slice. The identity $R_z^2/C = 1$ makes the formula valid at any fixed open-sheet height; no height-independent Fisher tensor is assumed. Parameterize the nontrivial edge by $p(t) = p_0 + tv$. Since $\nu \cdot p$ is constant, $\nu^{\mathsf{T}} \Sigma(p) \nu = \nu_l^2 l + \nu_m^2 m - (\nu \cdot p)^2$ is affine in t and strictly positive in the simplex interior. Meanwhile $r^2 = |p(t) - w|^2$ is a quadratic with positive leading coefficient. Their product has degree two or three and cannot be constant on an interval. Thus no common scale makes this full Fisher block compatible with the straight edge. This rejects the stronger identification for the stated model, not the angular-only theorem or a measured human discrimination tensor.

C Paired carriers and color-sensitive readout

The effect algebra alone does not determine how two paired state spaces encode color. Let $u : H_1 \to H_2$ be an isometry between complex Hilbert spaces, $P = uu^*$, and identify the active space $H_1 \oplus uH_1$ with $\mathbb{C}^2 \otimes H_1$. On it set $J = \sigma_z \otimes I$, $R = \sigma_x \otimes I$, $Q = \sigma_y \otimes I$, and $E = \frac{1}{2}[(1+\zeta)I + \xi R + \eta Q]$. For

$$V_\varphi \psi = 2^{-1/2}(\psi \oplus e^{i\varphi} u\psi) \tag{48}$$

direct multiplication gives

$$V_\varphi^* E V_\varphi = \tfrac{1}{2}(1 + \zeta + \xi \cos\varphi + \eta \sin\varphi) I. \tag{49}$$

For a fixed graph phase these effects have statistics independent of the normalized input ψ. Identifying hue with the sector phase would therefore require additional preparation or control data. The

indefinite form defined by J vanishes on the aligned graph, $V_0^* J V_0 = 0$; restricting that form does not supply a nondegenerate color metric.

Internal color-sensitive observables can instead be paired without changing the graph. For any effect $0 \le F \le I$ on H_1,

$$V_0^* \operatorname{diag}(F, uFu^*) V_0 = F. \tag{50}$$

These probabilities can vary with the input state. A stimulus encoding and a physically justified measurement family must still select F. An internal two-level family may itself have the effect bicone; its geometry does not identify hue with the sector coordinate.

Paired preparations also leave the readout on $\ker u^*$ undetermined. All effects $uFu^* + D$ with $0 \le D \le I - P$ have the same compression $u^*(uFu^* + D)u = F$. They remain indistinguishable for preparations and dynamics confined to uH_1; justified dynamics mixing that range with its complement can distinguish them. Static paired probabilities therefore do not assign chimerical colors to the complement, and a proper convex preparation image need not generate any such linear complement.

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Disclosures

The author is affiliated with Alpha Omega Lattice Corporation. No physiological or psychophysical measurements are reported in this article.

Data availability

Code and input data are available at `https://github.com/AlphaOmega-Institute/magenta-gap`. The numerical illustration uses the package in `recompute/`, preserved at commit `a0282ae`. The continuum certificate and its exact input are in `research/object-colour`; the finite effect-algebra checks are in `research/colour-interface`, preserved at commit `cb997e8`. The certificate input CSV has SHA-256

```
144fc58d2f3b01cd8ce5ae765199fd1a2af6
58b7bea7a540aa3cea6bb9d02c40
```

where the two lines concatenate. The arguments and assumptions used in this article are stated here; the code and rational enclosures permit reproduction of the specified computation.

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